



# Group H

# Application of CRISPR/Cas9 on CAR-T Therapy

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## Part 1 Introduction

The background of the slide features a faint, light blue DNA double helix structure. Overlaid on this is a close-up, slightly blurred image of a person's hand holding a glass pipette, with a small droplet of liquid being dispensed. The overall color palette is soft and scientific.

The number of cases of cancer in 2022 will increase by **32%** compared to 2007 [1]

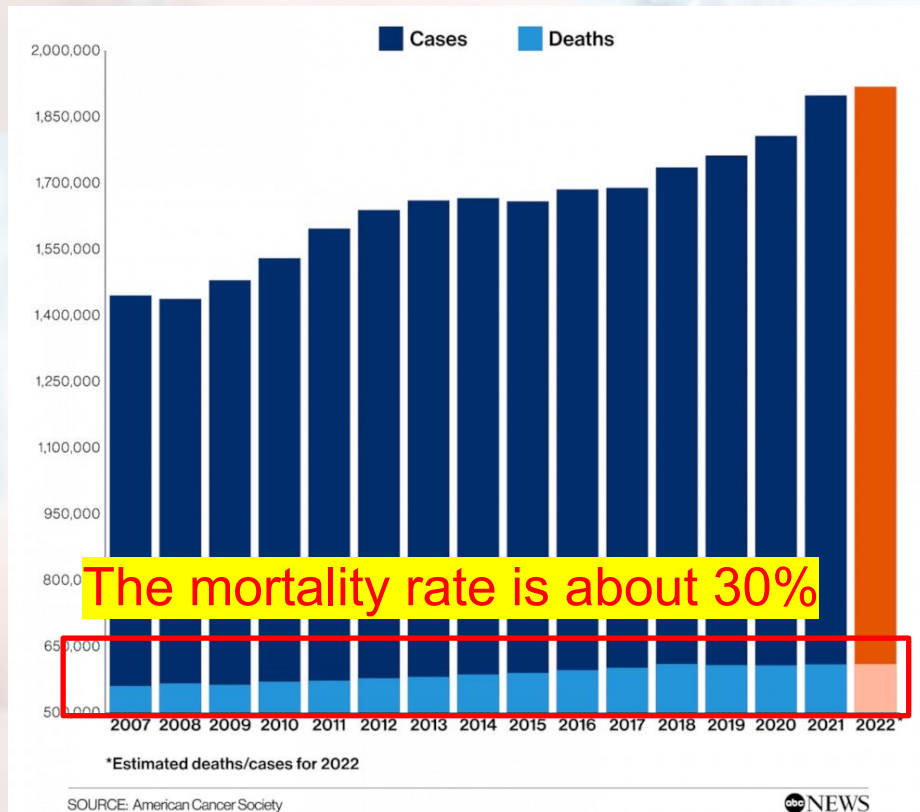


Figure.1 Estimated Cancer Cases and Death in the U.S.[1]

Numerous  
High mortality rate  
Dangerous



Figure.2 Cancer data in the United States[2]



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## Harmful



Damage normal cells

Vomiting

Low red blood cells

Possible to recur

and



Damage normal cells

Feeling tired

Hair loss

Possible to recur



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A new method



Figure.3 CAR-T therapy



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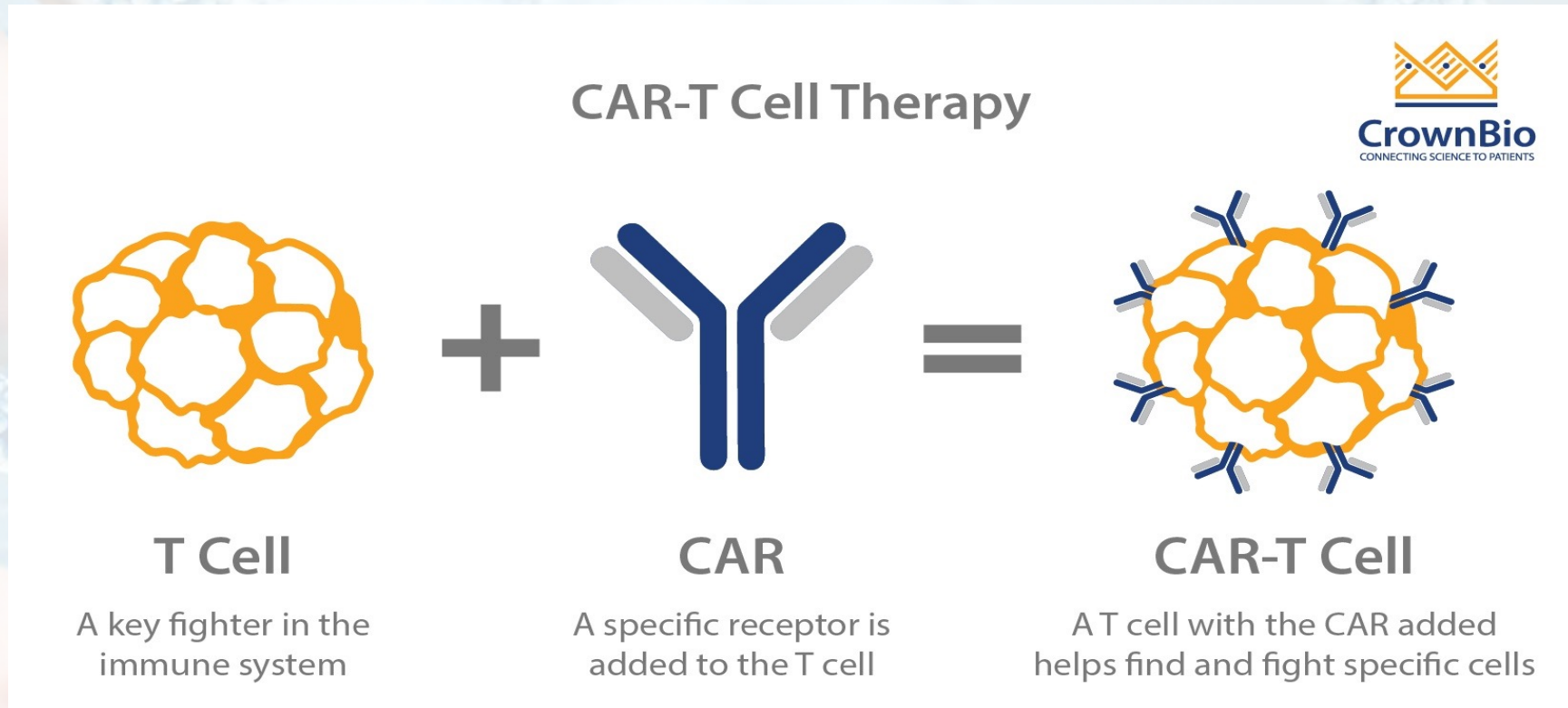


Figure 4. CAR-T Cell [3]

Therefore, CAR-T provides a low-side-effects and long-lasting therapy to cure cancer.



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## Part 2 Apply CRISPR/Cas9 on CAR-T

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CRISPR/Cas9 is a **gen-3** genome edition technology

high accuracy

High efficiency

Multifunction

Let CAR-T enter clinical application

## Components in CRISPR-Cas9

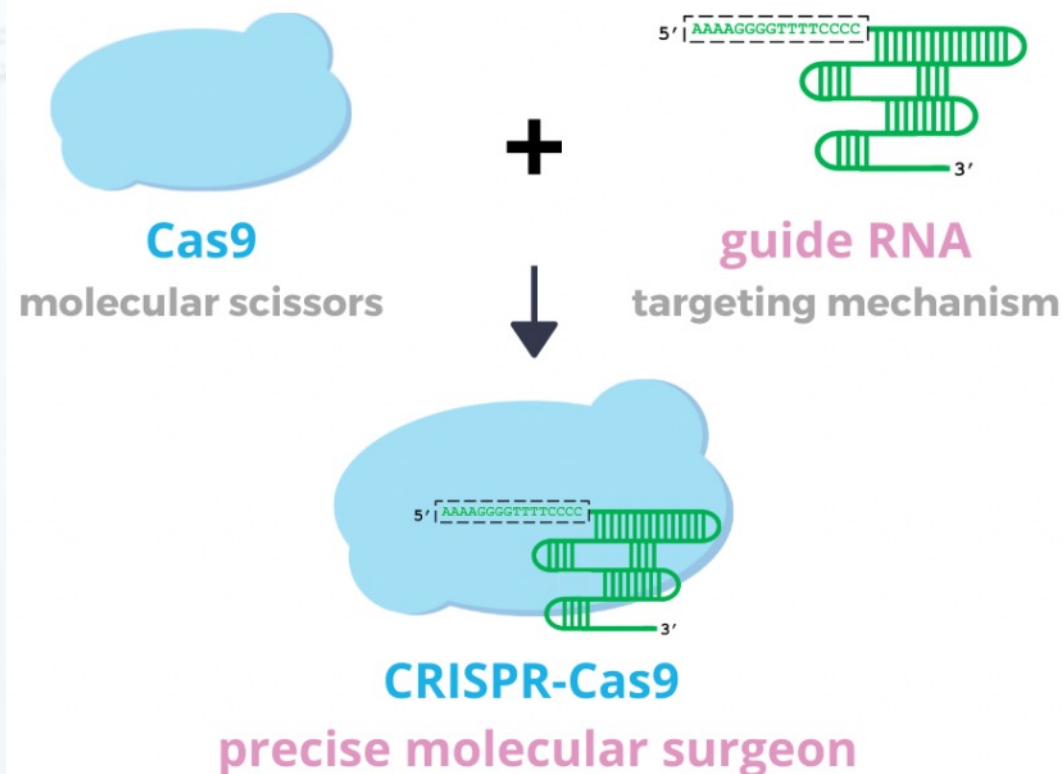


Figure 5. The structure of the CRISPR-Cas9[4]

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Immune checkpoints cause the death of T-cell

PD-1/PD-L1

CTLA-4/CD80(CD86)

TIM-3/CEACAM-1

GM-CSF factor

TGF $\beta$  factor

Stimulate the growth of tumor cell

CAR structure

Tumor antigen

Identify the tumor cell

Knock out the gene which stop T cell's killing and help the escape of tumor cell.

Modify the gene to enhance the killing efficiency of T cell. [4]

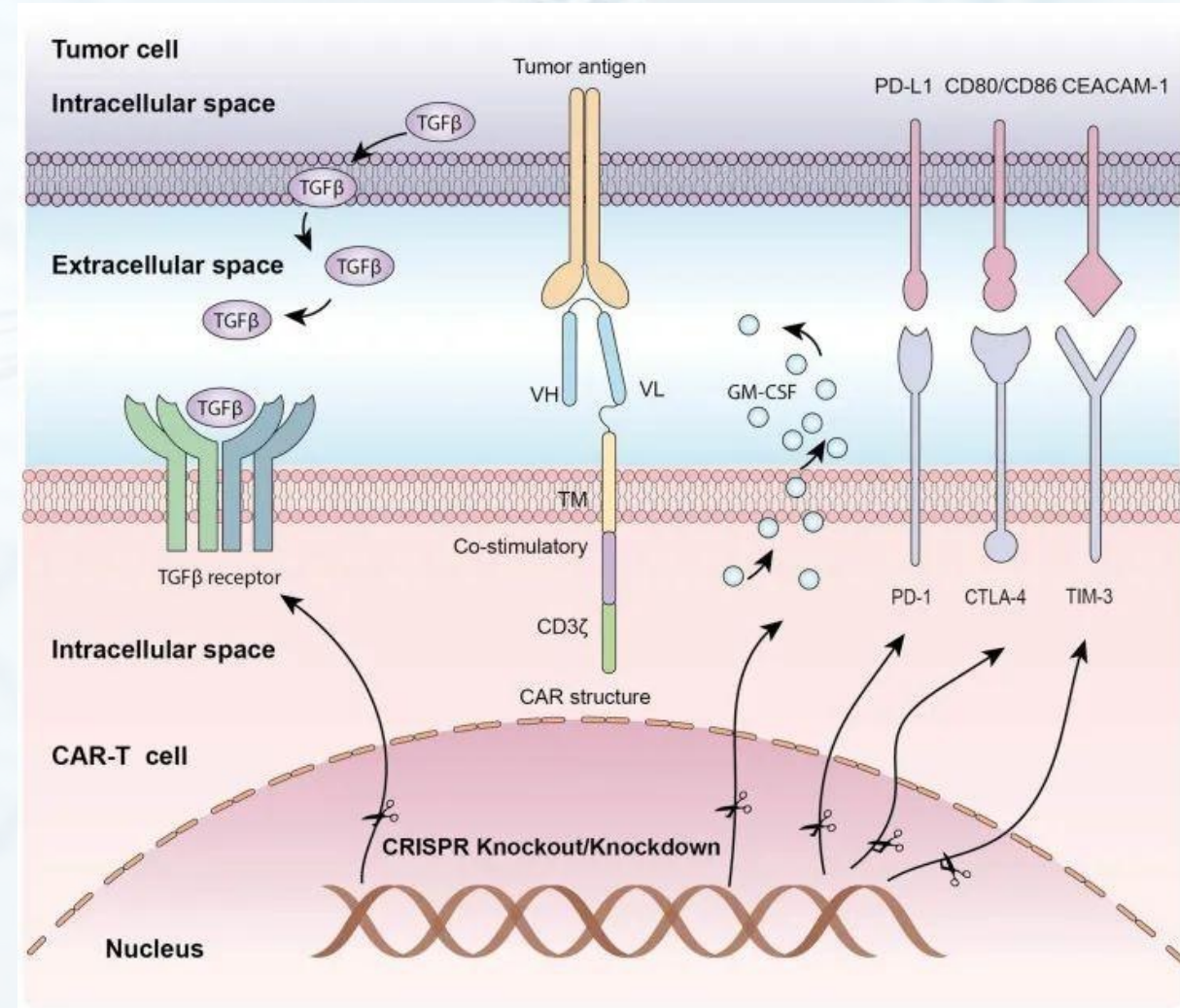


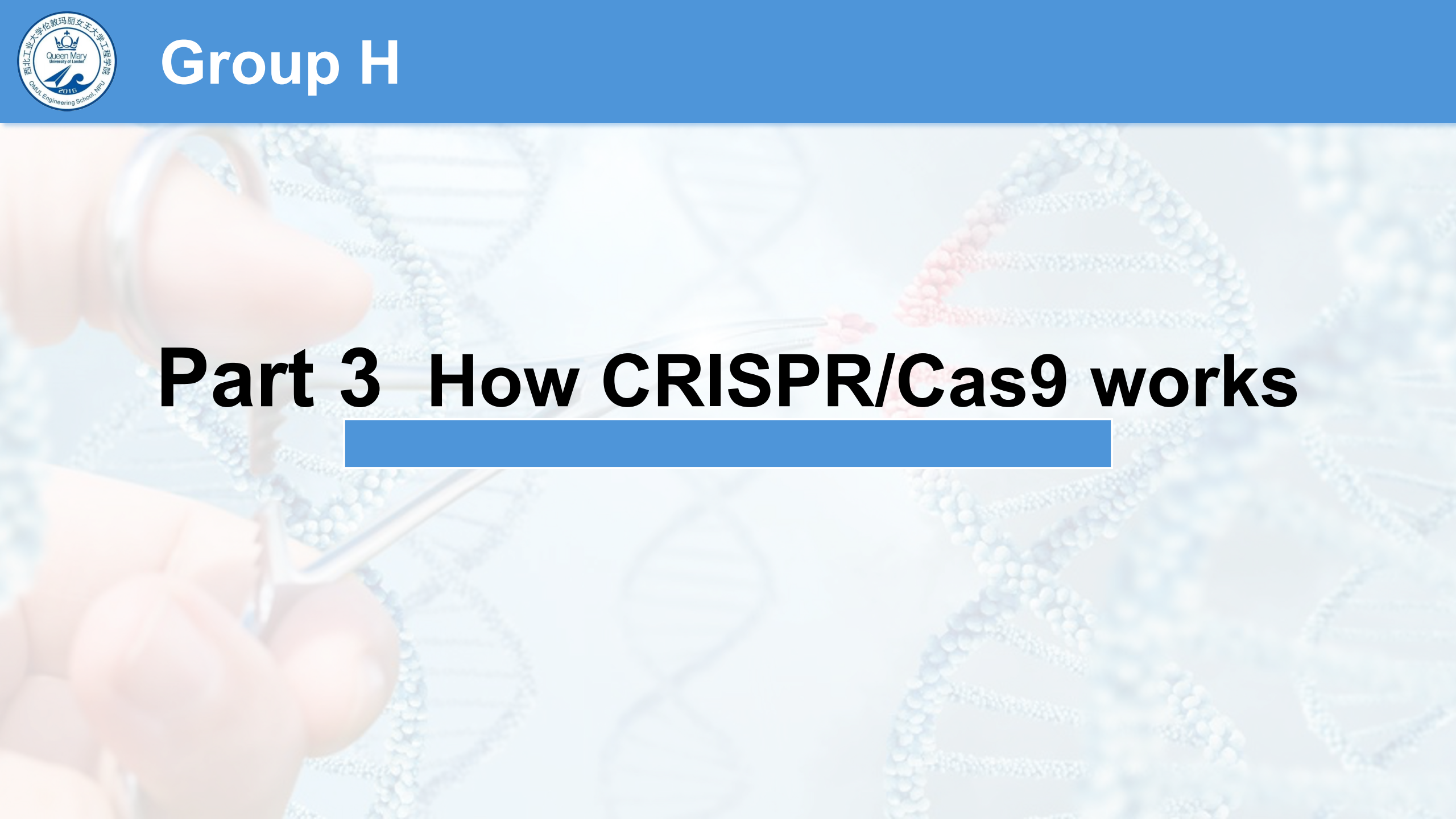
Figure.6 Application of CRISPR/Cas9 system in genome edition of CAR-T cell [4]





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## Part 3 How CRISPR/Cas9 works

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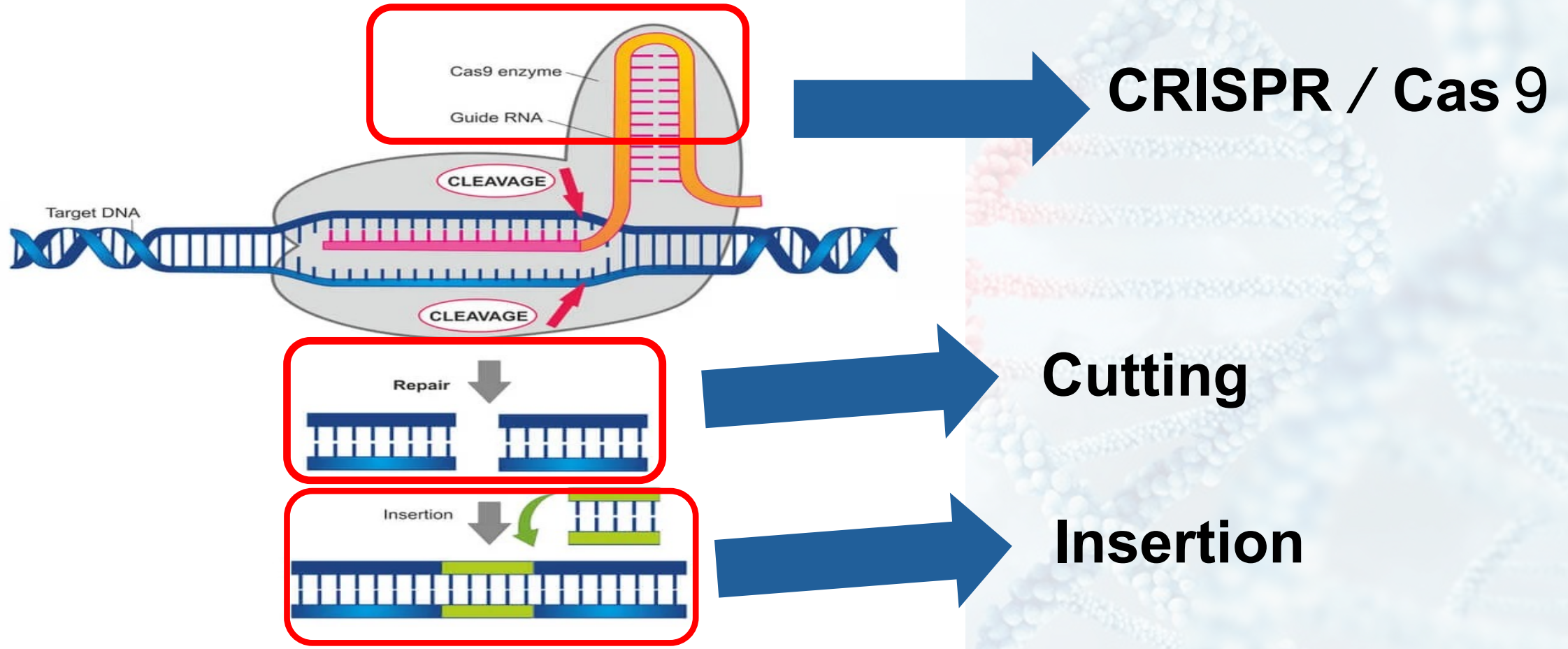


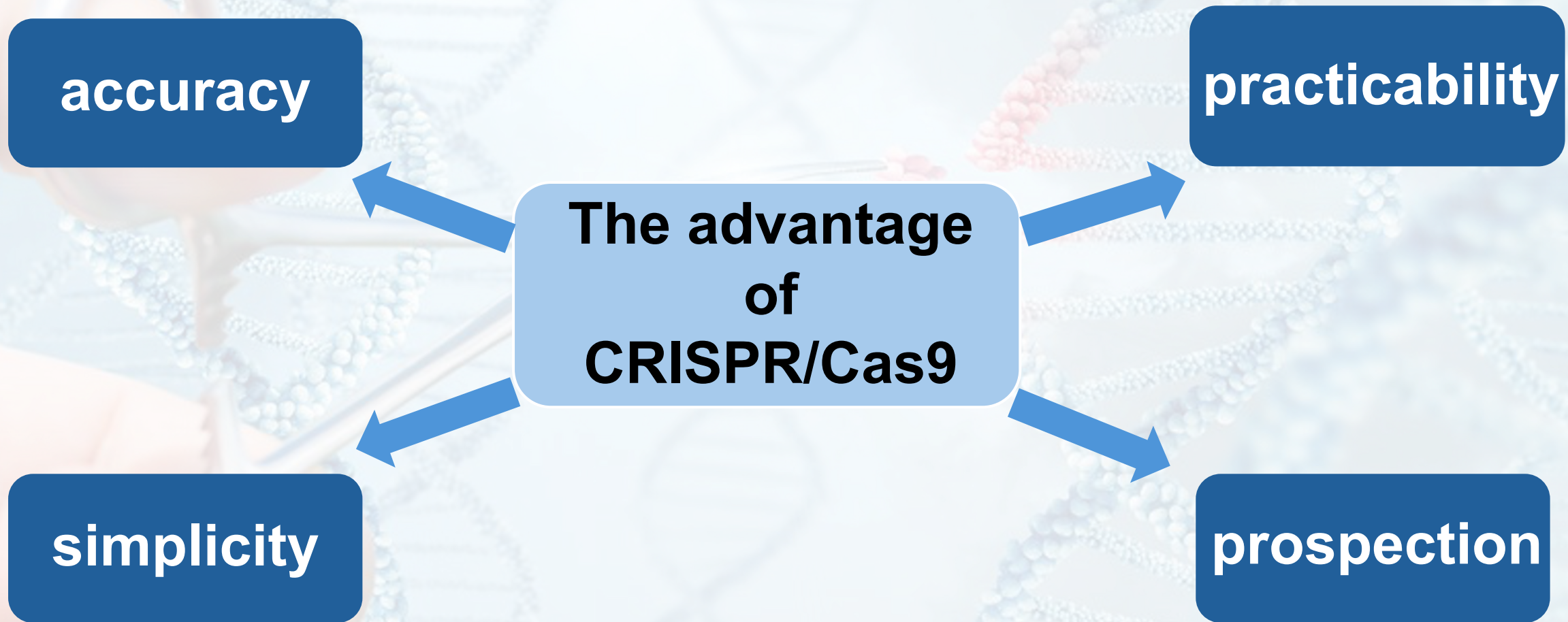
Figure 7. The principle of the CRISPR-Cas9[5]

Therefore, CRISPR/Cas9 play an important role in Car-T cell Therapy.





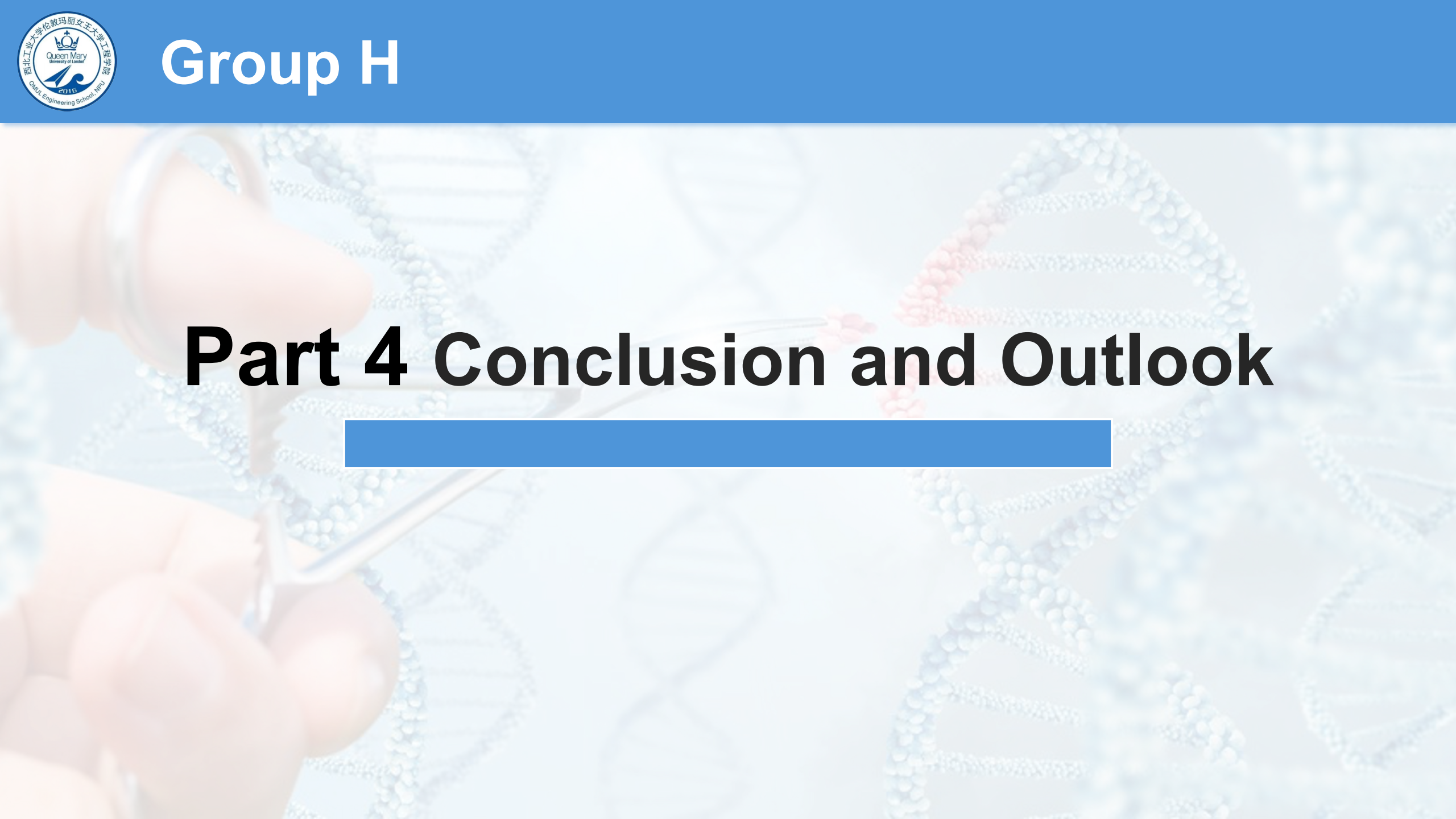
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## Part 4 Conclusion and Outlook

A faint, light blue background image of a DNA double helix structure, with a pair of hands holding a small portion of it in the center. A solid blue horizontal bar is positioned below the title text.





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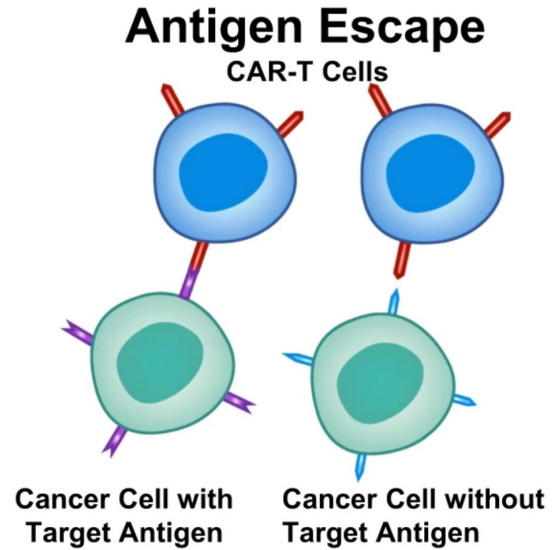
## Comparision

| Chemotherapy[6]        | Radiation Therapy[7]   | Targeted Medicine Therapy[8]         | CAR-T[9]                     |
|------------------------|------------------------|--------------------------------------|------------------------------|
| Non-selective hurting  | Non-selective hurting  | Only targeted on tumor cells         | Only targeted on tumor cells |
| Chemical based         | Radiation based        | Bio-centered chemical                | Immune cells from patient    |
| All cancer             | All cancer             | Specific cancers                     | Specific cancers             |
| Relative cheap         | Relative cheap         | Relative expensive                   | Very expensive               |
| Most accessible        | Most accessible        | Relative less accessible             | Difficult to access          |
| Relative bad prognosis | Relative bad prognosis | Optimistic prognosis                 | Very good prognosis          |
| Possible to recur      | Possible to recur      | Relatively less possibility to recur | Almost completely cure       |

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Toxic reaction



Antigen Escape

Figure 8. The cost of CAR T-cell therapy.[10]



Expensive

## Disadvantage





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## Expectations

1. Reduce the side-effect
2. Research for more common cancers
3. decrease the cost





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## Reference

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- [2] National Center for Chronic Disease Prevention and Health Promotion (NCCDPHP), Chronic Disease Fact Sheets, (<https://www.cdc.gov/chronicdisease/resources/publications/factsheets/cancer.htm>)
- [3] Ludovic Bourré, How to evaluate CAR T cell therapy in preclinic, (<https://www.crownbio.cn/2018-2-20/>)
- [4] Lester Quah, The CRISPR-Cas9 Genome Editing System, (<https://blog.advanxhealth.com/the-crispr-cas9-genome-editing-system/>)
- [5] Clara Rodríguez Fernández, CRISPR-Cas9: The Gene Editing Tool Changing the World, (<https://www.labiotech.eu/in-depth/crispr-cas9-review-gene-editing-tool/>)
- [6] Peter Nygren & Rolf Larsson. (2008) Predictive tests for individualisation of pharmacological cancer treatment. Expert Opinion on Medical Diagnostics 2:4, pages 349-360
- [7] Baskar R, Lee KA, Yeo R, Yeoh KW. Cancer and radiation therapy: current advances and future directions. Int J Med Sci. 2012;9(3):193-9. doi: 10.7150/ijms.3635. Epub 2012 Feb 27. PMID: 22408567; PMCID: PMC3298009.
- [8] Tsimberidou, AM. Targeted therapy in cancer. Cancer Chemother Pharmacol 76, 1113–1132 (2015). <https://doi.org/10.1007/s00280-015-2861-1>.
- [9] Sterner, R.C., Sterner, R.M. CAR-T cell therapy: current limitations and potential strategies. Blood Cancer J. 11, 69 (2021). <https://doi.org/10.1038/s41408-021-00459-7>
- [10] Gina Shaw, Hospitals Still Grappling With \$1 M+ Price Tag for CAR-T Rx, (<https://www.pharmacypracticenews.com/Clinical/Article/10-21/Hospitals-Still-Grappling-With-1-M-Price-Tag-for-CAR-T-Rx/64913?ses=ogst>)





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**Thank you for your listening**

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B. The gene express the factor induce of the death of tumor cell



A. The gene express the factor stimulate the growth of tumor cells

C. The gene express immune factor to call on other immune cells

## Knockout which one?



A. Gene modifies CAR structure



B. Gene produces digest enzyme



C. Gene related to cell division

## Insert which one?